

Q3 Network Bandwidth Expansion Plan

NETWORK CAPACITY AND BANDWIDTH EXPANSION PLAN

Reporting Period: Q3 2026 Prepared By: Department of Network Engineering Target Infrastructure: Seattle Data Center (Core Routing)

Overview: Over the past six months, inbound traffic to the Seattle Data Center has grown by an average of 18% month-over-month, largely driven by the deployment of the new high-resolution video rendering service. Current peak utilization of our dual 100 Gbps core uplinks frequently exceeds 85% between 18:00 and 22:00 PST. To prevent packet loss and maintain our 99.99% uptime SLA, proactive capacity expansion is required.

Current State Analysis: - Core Routers: 2x Nexus 9000 Series - Current Uplink Capacity: 200 Gbps (Active-Active LACP) - Peak Bandwidth Utilization (July 2026): 172 Gbps - Identified Bottlenecks: Edge firewall inspection throughput is plateauing at 80 Gbps per appliance.

Proposed Upgrades: 1. Uplink Expansion: We propose upgrading the dark fiber interconnects to the primary ISP from 100 Gbps to 400 Gbps QSFP-DD transceivers. This will quadruple our aggregate edge capacity to 800 Gbps.

2. Firewall Replacement: The legacy Layer 7 inspection firewalls will be replaced with next-generation clustered security appliances capable of 150 Gbps threat-prevention throughput each, deployed in a high-availability (HA) cluster.

3. Internal Top-of-Rack (ToR) Switches: Leaf switches in server rows 14 through 20 will be upgraded from 10G to 25G to accommodate the increased internal east-west traffic generated by the video rendering cluster.

Budget and Timeline: The total estimated capital expenditure (CapEx) for hardware procurement is \$415,000. Delivery of the new 400G transceivers and firewalls is expected by August 30. The maintenance window for the core uplink cutover is tentatively scheduled for Sunday, September 13, at 02:00 PST. A brief 15-minute degradation of external routing is anticipated during the BGP session convergence.